

Jiwon Kang

dryyrad0@snu.ac.kr — +82-10-8235-6003

Last updated: April 2026

Education

Seoul National University (SNU)

B.S. in Biosystems Engineering (Mechanical Track) & Artificial Intelligence

Mar 2023 – Present

- Full Scholarship (National Science & Engineering Scholarship)
- Expected Graduation: Aug 2027

Hana Academy Seoul (HAS)

High School Diploma

Mar 2019 – Feb 2022

Research Interests

Robotics, Reinforcement Learning, Motion Planning, Embodied AI, Vision-Language-Action (VLA)

Research Experience

KAIST Industrial & Systems Engineering

Research Intern

Jun 2025 – Oct 2025

- Developed a scene graph-based perception and reasoning framework for embodied robotic systems using RGB and LiDAR data
- Designed a ROS-based system integrating perception, navigation, and VLM-based decision making
- Proposed multimodal grounding method bridging 3D spatial representation with language reasoning
- Co-authored a research paper on scene graph grounded spatial understanding (under review at IEEE RA-L)

MIPAL (Machine Intelligence & Pattern Analysis Lab), SNU

Research Intern

Jan 2026 – Apr 2026

- Designed and evaluated approaches for VLA and non-VLA robotic control
- Developed a prototype social navigation system using multimodal perception (RGB + LiDAR)
- Explored hybrid architectures combining VLM-based reasoning with action policies

Learning Science Research Institute, SNU

Intern

Jun 2024 – Jul 2024

- Developed AI-based learning tools using generative models
- Designed and tested user-interactive educational applications

Projects

claw-use — Y Combinator MCP Apps Hackathon Finalist

International Hackathon

Jan 2026

- Built an MCP-based system for monitoring and controlling persistent OpenClaw agents via conversational interfaces
- Demonstrated improved usability and controllability of long-running autonomous agents
- Designed real-time agent observability and intervention without direct server access
- Enabled interactive steering of long-running autonomous agent workflows
- Developed and demonstrated a working prototype as a hackathon finalist

Transmission Design for Agricultural UTV (1st Prize)

Machine Elements in Mechanical Design Course

Apr 2025 - Jun 2025

- Designed and optimized a multi-stage transmission system under AGMA/ISO constraints

- Applied GA and SQP for constrained design optimization
- Performed strength analysis and validated system through 3D assembly

International AI Hackathon (SNU–UW–NTU)

Feb 2025

Participant

- Developed AI-based travel & restaurant order assistant using OCR and image generation
- Collaborated in international team environment

Experience

SNU Humanoid And Physical Embodied AI Club (SHAPE)

2026 – Present

Vice President & Physical AI Track Lead

- Lead Physical AI track, organizing research seminars on embodied AI and VLA
- Coordinate humanoid robotics projects and competition teams
- Bridge research and real-world robot system development

IROJE Patent & Law Firm

2024 – 2025

Trademark Design Consultant

- Advised 20+ companies on brand identity and technology positioning

Biosystems Engineering Department, SNU

2024 – 2025

Vice President

- Led student initiatives and coordinated academic programs

Awards

- National Scholarship for Excellence in Science and Technology (Full Ride)
- Scholarship for Excellence in Biosystems Engineering Activities (SNU)

Skills

Programming: Python, C++, MATLAB

Tools: ROS2, Isaac Sim, Mujoco, 3D CAD

Areas: Vision-Language-Action (VLA), Embodied AI, Robot Learning, Motion Planning